Functions

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C++

Inline Function

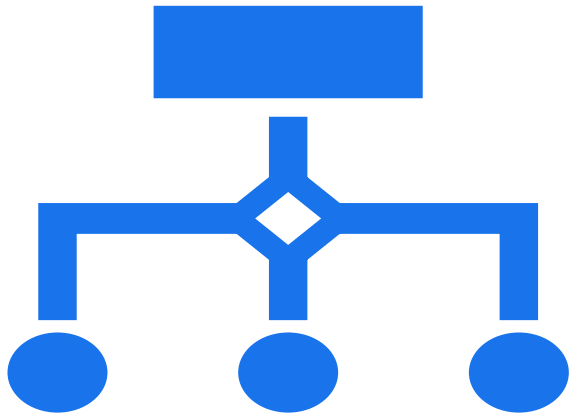
An **inline function** is a way of **telling the compiler** “Instead of calling this function like usual, please insert its code directly where it's used.”

It helps avoid the **performance cost** of jumping to a separate function, especially for **very short functions**.

Purpose: To reduce overhead of function calls in performance-critical code.

Example

```
inline int multiply(int a, int b) {
    return a * b;
}

int main() {
    int result = multiply(3, 4);  // Compiler replaces with: int result = 3 * 4;
    cout << result;
}
```



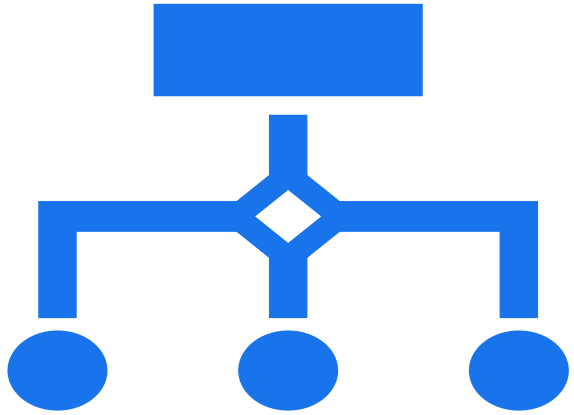
Why inline function

Function save memory space because all the calls to the function causes the same code to be executed , the function body need not be duplicated in body .

Compiler sees function call -> it jumps to function -> executes code inside function -> jump back to the instruction below calling code.

It saves memory space but it takes some extra time.

> For large code blocks, it's okay to use normal functions because saving memory is more important than a bit of speed. > But for small, short functions, using inline is better—because if we use a normal function instead, we lose speed just to save a tiny bit of memory. > In short: **use inline functions for short tasks to improve speed**, and **use regular functions for bigger tasks to save memory.**



Inline Functions

When to use:

- Short functions (1–2 lines).
- Frequently used functions.

When not to use:

- Recursive functions.
- Functions with loops or switch statements.
- Very large functions

Function Overloading

Function overloading means having **multiple functions with the same name** but **different parameter lists** (number, types, or both).

Purpose

- Improves code readability.
- Helps perform similar operations for different data types.

Example

```
int add(int a, int b) {
    return a + b;
}

float add(float a, float b) {
    return a + b;
}

int add(int a, int b, int c) {
    return a + b + c;
}
```

```
add(3, 4);        // Calls int version
add(3.5f, 2.5f);  // Calls float version
add(1, 2, 3);     // Calls 3-parameter version
```

Rules

- Same function name.
- Different number or type of parameters.
- Compiler selects the correct function based on arguments.

Invalid Overloads

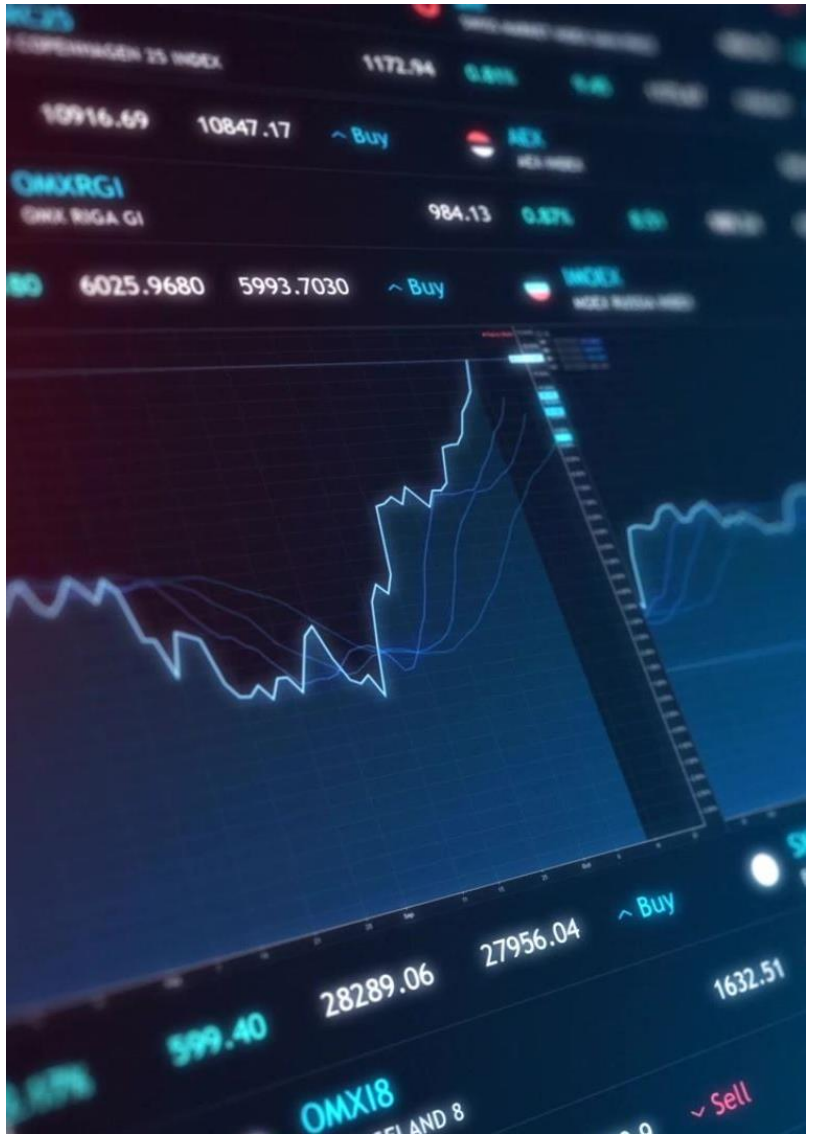
```
int show();

float show(); // ERROR: return type alone cannot distinguish overloads
```

Default Arguments

Default arguments allow **a function to be called with fewer arguments** than declared. The missing values are automatically filled with **default values**.

Syntax

void display(int a, int b = 10);



Example

```
#include <iostream>

using namespace std;

void display(int a, int b = 10, int c = 20) {
    cout << "a: " << a << ", b: " << b << ", c: " << c << endl;
}

int main() {
    display(1);        // b = 10, c = 20
    display(1, 2);     // c = 20
    display(1, 2, 3);  // All provided
    return 0;
}
```

Rules for Default Argument

Default values must be assigned **from right to left**.

You cannot skip a default argument in the middle.

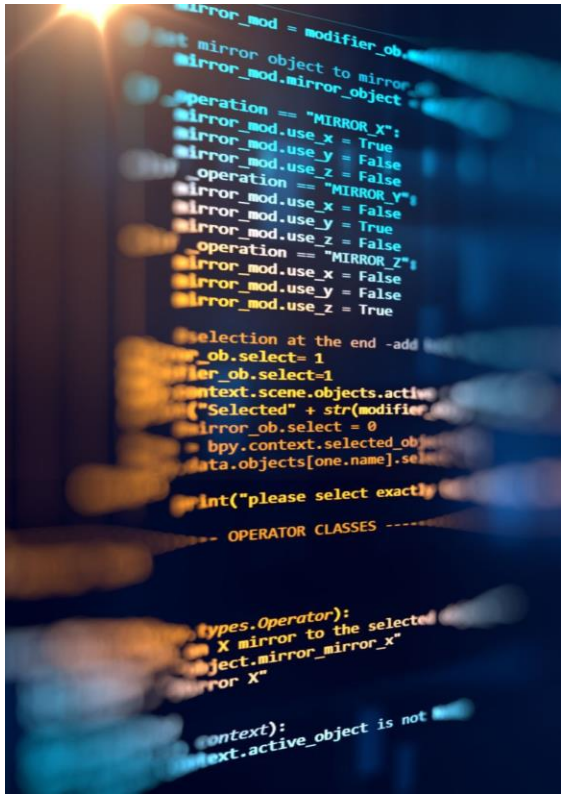
Correct

```
void example(int x, int y = 5, int z = 10);
```

Incorrect

```
void example(int x = 5, int y, int z = 10); // y does not have default
```

Pass by Reference (Reference Arguments)



Passing arguments by **reference** means the function works **on the original variables**, not on their copies.

void update(int &ref);

Reference arguments allow a function to modify the original variables by passing them by reference (using &).

Code

```
void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

int main() {
    int x = 5, y = 10;
    swap(x, y);
    cout << "x: " << x << ", y: " << y;
}
```

- Declared using the & symbol in the function parameter.
- No copy is made; the function works with the original variable.
- Useful for modifying values or improving performance.

When to Use:
When you want the function to **change the actual value**.
When you want to **avoid copying large data** (like arrays or objects).

Advantage

Saves memory and time.

Allows functions to **modify original data**.

Useful for large data structures (e.g., arrays, objects).

Pass By Value vs reference

Basis	Call by Value	Call by Reference
What is passed	A copy of the actual value	The address (reference) of the actual value
Changes in function	Do not affect the original variable	Affect the original variable
Memory usage	More (copies are created)	Less (no copies, direct reference)
Safety	Safer, original data is protected	Riskier, original data can be modified

Return by Reference

- Returning by reference means a function returns the address (reference) of a variable instead of just its value. This allows the caller to directly access and modify the original variable through the returned reference.

When to use

- You need to modify an existing variable directly.
- Avoiding unnecessary copies of large objects for efficiency.
- Implementing operator[] or operator= in custom classes.

Example

```
#include <iostream>

using namespace std;

int x = 5;  // Global variable

int& give() {

    return x;  // Returning reference to global
variable   , return address

}

int main() {

    give() = 100;  // Changes x directly

    cout << x;     // Output: 100

    return 0;

}
```

Recursive Function

- A recursive function in C++ is a function that calls itself to solve smaller instances of a problem. Recursion is widely used in problems like factorial computation, Fibonacci series, and tree traversal.

```
#include <iostream>
using namespace std;

int factorial(int n) {
    if (n <= 1) // Base case
        return 1;
    else
        return n * factorial(n - 1); // Recursive call
}

int main() {
    int num = 5;
    cout << "Factorial of " << num << " is: " << factorial(num) << endl;
    return 0;
}
```